



Indian Institute of Technology Ropar
Department of Mathematics
MA302 - Optimization Methods with Applications
Second Semester of the Academic Year 2025-26

Tutorial sheet - 3

1. Find all the basic solutions of the following problem:

Maximize

$$Z = x_1 + 3x_2 + 3x_3$$

subject to

$$x_1 + 2x_2 + 3x_3 = 4$$

$$2x_1 + 3x_2 + 5x_3 = 7$$

Also, find which of the basic feasible solutions are:

- (a) Basic feasible,
 - (b) Non-degenerate basic feasible, and
 - (c) Optimal basic feasible,
2. Is the following solution

$$x_1 = 1, \quad x_2 = \frac{1}{2}, \quad x_3 = x_4 = x_5 = 0$$

a basic solution of the equations

$$x_1 + 2x_2 + x_3 + x_4 = 2$$

$$x_1 + 2x_2 + \frac{1}{2}x_3 + x_5 = 2$$

3. Solve the LPP using simplex method:

Maximize

$$Z = 3x_1 + 4x_2$$

subject to

$$x_1 + x_2 \leq 450$$

$$2x_1 + x_2 \leq 600$$

where

$$x_1, x_2 \geq 0$$

4. Solve the LPP using simplex method:

Maximize

$$Z = 4x_1 + 3x_2 + 6x_3$$

subject to

$$2x_1 + 3x_2 + 2x_3 \leq 440$$

$$4x_1 + 3x_3 \leq 470$$

$$2x_1 + 5x_2 \leq 430$$

where

$$x_1, x_2, x_3 \geq 0$$

5. Solve the LPP using simplex method and check whether the system has an unbounded solution.

Minimize

$$Z = x_1 - 10x_2$$

subject to

$$x_1 - 5x_2 \geq 2$$

$$-x_1 + 5x_2 \leq 5$$

where

$$x_1, x_2 \geq 0$$

6. Is the following a LPP?

Maximize

$$z = 4x_1 + 3x_2$$

subject to

$$x_1 + x_2 \leq 8 \text{ or } 2x_1 + x_2 \leq 10$$

where

$$x_1, x_2 \geq 0$$

7. Use the two-phase method to solve the following LPP.

a). **Maximize** $Z = 2x_1 + x_2 + \frac{1}{4}x_3$

Subject to constraints

$$4x_1 + 6x_2 + 3x_3 \leq 8$$

$$3x_1 - 6x_2 - 4x_3 \leq 1$$

$$2x_1 + 3x_2 - 5x_3 \geq 4$$

$$x_1, x_2, x_3 \geq 0$$

b). **Minimize** $Z = x_1 - 2x_2 - 3x_3$

Subject to constraints

$$-2x_1 + x_2 + 3x_3 = 2$$

$$2x_1 + 3x_2 + 4x_3 = 1$$

$$x_1, x_2, x_3 \geq 0$$